Vitor Farias Costa de Carvalho

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Education

Purdue University
Master of Science in Industrial Engineering (with thesis), GPA: 3.86/4.0 — 08/2023 – 05/2025
Bachelor of Science in Industrial Engineering, minor: Mathematics, GPA: 3.74/4.0 — 08/2019 – 05/2023

Skills

Programming/Tools: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, PySpark), R, SQL, MATLAB, Power BI, Excel, Git
Languages: English (full professional fluency), Portuguese (native), Spanish (proficient reading/writing; conversational)

Work Experience

Lafayette Urban Ministry (LUM) — Lafayette, IN
Data Analytics Intern — 10/2025 – 12/2025
- Built a data pipeline in Python (Pandas) to convert WordPress form submissions into standardized CSV files for Bloomerang imports, automating onboarding of existing volunteers into the CRM database
- Improved Bloomerang database integrity by identifying, correcting, and merging inconsistent volunteer profiles
- Developed a step-by-step instructional guide for migrating volunteer opportunities from LUM's website to the Bloomerang Volunteer platform, streamlining hour tracking, volunteer check-in/out, and administrative workflow

Purdue University — West Lafayette, IN
Graduate Researcher at CONNplexity Lab – Data Science/Machine Learning — 08/2023 – 05/2025
- Developed a novel data-driven framework to identify individuals from brain imaging data, similar to a biological fingerprint
- Validated approach on dataset containing 426 participants, achieving up to 100% identification accuracy and outperforming traditional and previously published methods
- Led the project end-to-end, from methodology design and analysis to interpretation and manuscript preparation
- Published the work as sole first author and presented findings at international conferences (https://doi.org/10.3390/app15094821)

Graduate Researcher at Programmable Structures Lab – Data Science/Machine Learning — 08/2023 – 05/2025
- Developed object-classification models using tactile data collected from piezoelectric sensor–embedded substrates
- Designed a sparse sensing framework to optimize sensor selection, achieving 90% classification accuracy while reducing sensor usage by 60%
- Applied signal-processing methods to reconstruct object shapes from raw tactile measurements

Programming Lab Instructor — 08/2023 – 12/2023
- Taught weekly lab sessions to 25 undergraduate students, covering fundamentals of SQL, R, HTML, CSS, JavaScript, and PHP
- Led live coding demonstrations, assisted students with lab/homework inquiries, and graded programming projects

Centro de Desenvolvimento de Sistemas — Brasília, DF
Software Engineering Intern — 05/2023 – 08/2023
- Optimized the front-end of Django application by implementing tables, menus, filters, and pagination, enabling faster client-side rendering and reducing page load times
- Enhanced back-end architecture by improving views and URL routing, streamlining database interactions through ORM, and implementing robust form and error handling processes
- Refactored legacy code using Git, fixing 150+ bugs and improving code quality and documentation

Projects

Song Recommender System (Python, Databricks)
- Processing a 1.2 million-song Spotify metadata dataset using Python (PySpark) in Databricks
- Designing a Bronze-Silver transformation pipeline to clean and structure tracks' features
- Integrating Last.fm API data to augment track metadata with user listening information

Personal Finance Dashboard (Power BI, Excel, Power Automate)
- Implemented an automated pipeline using Power Automate to parse incoming credit card transaction emails, extract transaction date, merchant, and amount, and load them into centralized Excel dataset
- Built merchant classification system in Excel leveraging XLOOKUP and custom mapping table to automatically assign spending categories (e.g., Food & Drink, Groceries, Shopping, etc.) to each transaction
- Created an interactive mobile Power BI dashboard with DAX measures and automated daily refresh to spending patterns by category and merchant through dynamic measures and visualizations

PepsiCo Fleet MPG Optimization (Python, Machine Learning)
- Conducted regression-based analysis and trained Random Forest models in Python (Pandas, NumPy, Scikit-learn) to identify key factors affecting fuel efficiency
- Translated analytical results into business recommendations for optimizing fleet allocation, driving an estimated 20% reduction in operational costs while directly supporting PepsiCo's Pep+ sustainability goals

COVID-19 Web Platform (R, SQL, Machine Learning)
- Collaborated on developing a respiratory disease diagnosis website application with secure user authentication, MySQL-based data management, symptom-driven illness prediction (allergies, cold, flu, COVID), and integrated user feedback features
- Applied a KNN classifier with k-fold cross-validation to evaluate model performance, and employed stratified bootstrapping to generate balanced resamples for illness prediction under class imbalance